

System Demonstration

```
Starting API with RAG Engine...
$ Initializing Dynamic RAG Engine...
Initializing embeddings model...
D:\Personals\agentic\agentic_rag\app\db\vector_store.py:9: LangChainDeprecationWarning: The class 'HuggingFaceEmbeddings' was deprecated in LangChain 0.2.2 and will be removed in 1.0. An updated version of the class exists in the 'langchain-huggingface' package and should be used instead. To use it run 'pip install -U 'langchain-huggingface'' and import as 'from langchain_huggingface import HuggingFaceEmbeddings'.
  self.embeddings = HuggingFaceEmbeddings(
D:\Personals\Project1\agentic_rag\venv\lib\site-packages\torch\utils\generic.py:311: FutureWarning: 'torch.utils._pytree._register_pytree_node' is deprecated. Please use 'torch.utils._pytree.register_pytree_node' instead.
  torch.utils._pytree._register_pytree_node(
Initializing vector store...
D:\Personals\agentic\agentic_rag\app\db\vector_store.py:13: LangChainDeprecationWarning: The class 'Chroma' was deprecated in LangChain 0.2.9 and will be removed in 1.0. An updated version of the class exists in the 'langchain-chroma' package and should be used instead. To use it run 'pip install -U 'langchain-chroma'' and import as 'from langchain_chroma import Chroma'.
  self.vector_store = Chroma(
[+] Dynamic RAG Engine ready!
RAG Engine initialized successfully
Initializing embeddings model...
Initializing vector store...
API ready!
INFO: Started server process [8320]
INFO: Waiting for application startup.
INFO: Application startup complete.
```

Fig 1: FastAPI Server Startup Logs Showing RAG Engine Initialization

```
StatusCode      : 200
StatusDescription : OK
Content         : {"status":"healthy"}
RawContent      : HTTP/1.1 200 OK
                  Content-Length: 20
                  Content-Type: application/json
                  Date: Fri, 27 Feb 2026 17:19:38 GMT
                  Server: uvicorn

                  {"status":"healthy"}
Forms           : {}
Headers         : [[Content-Length, 20], [Content-Type, application/json], [Date, Fri, 27 Feb 2026 17:19:38 GMT], [Server, uvicorn]]
Images          : {}
InputFields     : {}
Links           : {}
ParsedHtml      : mshtml.HTMLDocumentClass
RawContentLength : 20
```

Fig 2: Health Check API Response Showing Server Status

Agentic RAG API

1.0.0 OAS 3.1

/openapi.json

RAG system with ReAct agent using Hugging Face models

default			^
POST	/api/v1/upload	Upload Document	v
POST	/api/v1/query	Query	v
POST	/api/v1/chat	Chat	v
GET	/api/v1/documents	List Documents	v
DELETE	/api/v1/clear	Clear Database	v
GET	/	Root	v
GET	/health	Health Check	v

Fig 3: FastAPI Swagger UI Documentation Showing All Available Endpoints

Schemas			^
Body_upload_document_api_v1_upload_post	>	Expand all	object
ChatRequest	>	Expand all	object
ChatResponse	>	Expand all	object
DocumentInfo	>	Expand all	object
HTTPValidationError	>	Expand all	object
QueryRequest	>	Expand all	object
QueryResponse	>	Expand all	object
ReasoningStep	>	Expand all	object
Source	>	Expand all	object
UploadResponse	>	Expand all	object
ValidationError	>	Expand all	object

Fig 4: FastAPI Pydantic Schemas Showing Request/Response Data Models

```

• (venv) PS D:\Personals\agentic\agentic_rag> curl.exe http://localhost:8000/
{"message":"Agentic RAG API","version":"1.0.0","endpoints":["/api/v1/upload","/api/v1/query","/api/v1/chat","/api/v1/documents","/api/v1/clear"]}
• (venv) PS D:\Personals\agentic\agentic_rag> start http://localhost:8000/docs
• (venv) PS D:\Personals\agentic\agentic_rag> Get-Content sample.txt -Head 5

# PHYSICS

## Classical Mechanics
Newton's Laws of Motion form the foundation of classical mechanics. The first law states that an object at rest stays at rest and an object in motion stays in motion unless acted upon by a force. The second law defines force as mass times acceleration ( $F = ma$ ). The third law states that for every action, there is an equal and opposite reaction.
• (venv) PS D:\Personals\agentic\agentic_rag> curl.exe -X POST "http://localhost:8000/api/v1/upload" -F "file=@sample.txt"
{"filename":"sample.txt","chunks":14,"document_id":"bf563d12-4edb-49b9-9925-cc153417e393","timestamp":"2026-02-27T23:05:42.274359"}
• (venv) PS D:\Personals\agentic\agentic_rag>

```

Fig 5: Command Line Interface Showing API Root Endpoint, Document Upload, and File Content Preview

```

• (venv) PS D:\Personals\agentic\agentic_rag> curl.exe http://localhost:8000/api/v1/documents
[{"filename":"sample.txt","upload_date":"2026-02-26T13:47:57.618097","chunks":84,"document_id":"526d7f6e-84d8-5c22-9741-a171517df7b5"}]
• (venv) PS D:\Personals\agentic\agentic_rag> Get-Content Test.json | ConvertFrom-Json | ConvertTo-Json
{
  "question": "What are Newton's laws of motion?",
  "session_id": "test"
}
• (venv) PS D:\Personals\agentic\agentic_rag> curl.exe -X POST "http://localhost:8000/api/v1/query" -H "Content-Type: application/json" -d `@Test.json
{"answer":"# PHYSICS\n\n## Classical Mechanics\n\nNewton's Laws of Motion form the foundation of classical mechanics.", "sources":[{"document":"sample.txt","page":0,"chunk":"# PHYSICS\n\n## Classical Mechanics\n\nNewton's Laws of Motion form the foundation of classical mechanics. The first law states that an object at rest stay..."}], "reasoning_trace":[{"step":1,"thought":"Analyzing document structure","action":"document_analysis","observation":"Found 15 chunks, identified 10 sections"}, {"step":2,"thought":"Extracted exact answer sentence","action":"sentence_extraction","observation":"Match score: 3"}, {"step":3,"thought":"Retrieval complete","action":"document_search","observation":"Selected best match from document"}, {"step":4,"thought":"Returning answer","action":"respond","observation":"Answer ready"}], "retrieval_used":true}
• (venv) PS D:\Personals\agentic\agentic_rag>

```

Fig 6: Document Listing, JSON Query File, and RAG Query Response with Reasoning Trace

```
= *60
SCIENCE: How fast does light travel?
= *60
ANSWER:
Light is an electromagnetic wave traveling at 299,792 kilometers per second in a vacuum.

RETRIEVAL: True
SOURCE: sample.txt
- *60

= *60
SCIENCE: What is quantum mechanics?
= *60
ANSWER:
## Quantum Mechanics
Quantum mechanics describes the behavior of matter at atomic scales.

RETRIEVAL: True
SOURCE: sample.txt
```

Fig 7: Sample Science Questions with Correct RAG Responses Showing Retrieval Success

```
= *60
SCIENCE: What is the Milky Way?
= *60
ANSWER:
The Milky Way is a spiral galaxy containing over 100 billion stars.

RETRIEVAL: True
SOURCE: sample.txt
- *60

= *60
SCIENCE: When did humans first land on the Moon?
= *60
ANSWER:
## Space Exploration
The first human landed on the Moon in 1969.

RETRIEVAL: True
SOURCE: sample.txt
- *60
```

Fig 8: Additional Science Questions Demonstrating Consistent Retrieval